

Question 1

Find

$$\lim_{x \rightarrow a} \frac{x^x - x^a}{a^x - a^a}, \quad a > 0, a \neq 1.$$

Put $x = a$:

$$\frac{a^a - a^a}{a^a - a^a} = \frac{0}{0},$$

L'Hôpital: if $\frac{0}{0}$, then $\lim \frac{f}{g} = \lim \frac{f'}{g'}$

so L'Hôpital's rule can be used.

For $y = x^x$:

$$\ln y = x \ln x \Rightarrow \frac{y'}{y} = \ln x + 1,$$

Log differentiation: $(\ln y)' = \frac{y'}{y}$;
product rule: $(uv)' = u'v + uv'$

so

$$(x^x)' = x^x(\ln x + 1).$$

$$(x^a)' = a x^{a-1}$$

Power rule: $(x^n)' = nx^{n-1}$

$$(a^x)' = a^x \ln a, \quad (a^a)' = 0$$

Exponential rule: $(a^x)' = a^x \ln a$;
derivative of a constant = 0

$$\lim_{x \rightarrow a} \frac{x^x(\ln x + 1) - a x^{a-1}}{a^x \ln a}$$

L'Hôpital applied

Put $x = a$:

$$\frac{a^a(\ln a + 1) - a \cdot a^{a-1}}{a^a \ln a}.$$

Index law: $a^m a^n = a^{m+n}$

Since

$$a \cdot a^{a-1} = a^a,$$

we obtain

$$\begin{aligned} & \frac{a^a(\ln a + 1) - a^a}{a^a \ln a} \\ &= \frac{a^a \ln a + a^a - a^a}{a^a \ln a} = \frac{a^a \ln a}{a^a \ln a} = \boxed{1} \end{aligned}$$

Cancel common factor

Numerical check ($a = 2$):

$$f(x) = \frac{x^x - x^2}{2^x - 4}.$$

The values approach 1 as $x \rightarrow 2$:

x	1.9	1.99	2.01	2.1
$f(x)$	0.838	0.983	1.017	1.183

The values from both sides approach 1.

Question 2

Linear independence

Vectors $v_1, \dots, v_n$ are linearly independent if

$$c_1 v_1 + \dots + c_n v_n = 0 \quad \Rightarrow \quad c_1 = \dots = c_n = 0.$$

In simple words, none of the vectors can be written as a combination of the others.

Importance of linear independence:

1. It removes redundancy.

If one feature can be written using other features, it does not provide a new direction or new information.

For example, if

$$v_3 = 2v_1 + v_2,$$

then v_3 is redundant.

2. It helps us get a unique solution.

In linear regression, if the feature columns are linearly independent, then

$$X^T X$$

is invertible and the least-squares solution is uniquely determined:

$$\hat{w} = (X^T X)^{-1} X^T y.$$

If the features are dependent, $X^T X$ can be singular, so the weights may not be unique.

3. It tells us the true number of directions in the data.

The number of linearly independent vectors gives the dimension of the space. Dependent vectors do not increase the dimension.

4. It can make learning more stable.

When features are highly dependent, small changes in the data can cause large changes in the estimated weights. Removing redundancy can make the model more stable.

5. Connection to deep learning.

A neuron takes a weighted sum of its inputs:

$$z = w_1 x_1 + \dots + w_n x_n + b = \mathbf{w}^T \mathbf{x} + b.$$

This is a linear combination of the input features. If the features are independent, they can provide different information to the network instead of repeating the same information.

Main idea:

Linear independence means each vector contributes something new.

In machine learning, this matters because it reduces redundant features, helps give unique model parameters, and tells us how many independent directions are actually present in the data.

Question 3

$$E[u] = \sum_{i=1}^{N-1} (u_i - u_{i-1})^2, \quad u = (u_0, \dots, u_{N-1}).$$

We want $\nabla_u E$, so first we find $\frac{\partial E}{\partial u_k}$ for one variable u_k .

Independent vectors give new directions

Independent columns $\Rightarrow X^T X$ is invertible

dim = rank

Less multicollinearity $\Rightarrow$ more stable weights

Neural network: weighted sum = linear combination

1. Expand the sum.

Gradient = vector of all $\frac{\partial E}{\partial u_k}$

$$E[u] = (u_1 - u_0)^2 + (u_2 - u_1)^2 + \cdots + (u_{N-1} - u_{N-2})^2.$$

Take one variable u_k . For an interior point, it appears in two neighbouring terms:

$$(u_k - u_{k-1})^2 \quad \text{and} \quad (u_{k+1} - u_k)^2.$$

2. First term: $i = k$.

General term:

$$(u_i - u_{i-1})^2.$$

Substitute $i = k$

To make u_k the first variable, put $i = k$:

$$(u_i - u_{i-1})^2 \rightarrow (u_k - u_{k-1})^2.$$

3. Second term: $i = k + 1$.

To make u_k the second variable, we need

Substitute $i = k + 1$

$$i - 1 = k,$$

so

$$i = k + 1.$$

Hence,

$$(u_i - u_{i-1})^2 \rightarrow (u_{k+1} - u_k)^2.$$

4. Differentiate.

Sum rule; derivative of a constant = 0

$$E = (u_k - u_{k-1})^2 + (u_{k+1} - u_k)^2 + \text{terms without } u_k.$$

Terms without u_k are constants with respect to u_k , so their derivatives are 0.

Therefore,

$$\frac{\partial E}{\partial u_k} = \frac{\partial}{\partial u_k}(u_k - u_{k-1})^2 + \frac{\partial}{\partial u_k}(u_{k+1} - u_k)^2.$$

First part:

$$\frac{\partial}{\partial u_k}(u_k - u_{k-1})^2 = 2(u_k - u_{k-1}).$$

Chain rule: $\frac{d}{dx}(f^2) = 2f f'$

Second part:

$$\frac{\partial}{\partial u_k}(u_{k+1} - u_k)^2 = -2(u_{k+1} - u_k).$$

The minus sign occurs because

$$\frac{\partial}{\partial u_k}(u_{k+1} - u_k) = -1.$$

$$\frac{\partial E}{\partial u_k} = 2(u_k - u_{k-1}) - 2(u_{k+1} - u_k)$$

Simplify: $2u_k + 2u_k = 4u_k$

Therefore,

$$\boxed{\frac{\partial E}{\partial u_k} = 2(2u_k - u_{k-1} - u_{k+1})}, \quad 1 \leq k \leq N - 2.$$

5. Boundary points.

u_0 appears only in

$$(u_1 - u_0)^2.$$

Thus,

$$\frac{\partial E}{\partial u_0} = 2(u_0 - u_1).$$

Similarly, u_{N-1} appears only in

$$(u_{N-1} - u_{N-2})^2,$$

so

$$\frac{\partial E}{\partial u_{N-1}} = 2(u_{N-1} - u_{N-2}).$$

6. Final gradient.

$E = \|Du\|^2$, D = difference matrix

$$\nabla E = \begin{bmatrix} 2(u_0 - u_1) \\ 2(2u_1 - u_0 - u_2) \\ \vdots \\ 2(2u_{N-2} - u_{N-3} - u_{N-1}) \\ 2(u_{N-1} - u_{N-2}) \end{bmatrix}.$$

In matrix form,

$$\boxed{\nabla E = 2D^T Du}.$$

Reflection: At first I saw the gradient only as a list of partial derivatives to compute. Later I understood that it points in the direction of steepest increase, its magnitude tells how rapidly the function changes, and moving against it reduces the function. This is the basis of optimization and learning.

Question 4

Given

$$X \sim \text{Poisson}(\lambda)$$

with PMF

$$P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

X represents the number of events occurring in a fixed interval, while λ represents the average rate of occurrence.

(a) Derive $E[X]$

$$E[X] = \sum_{x=0}^{\infty} x \frac{e^{-\lambda} \lambda^x}{x!}.$$

Definition of expectation

The $x = 0$ term is 0. For $x \geq 1$,

$$x! = x(x-1)!$$

$$\frac{x}{x!} = \frac{1}{(x-1)!}.$$

Therefore,

$$E[X] = e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}.$$

Let

$$k = x - 1.$$

Change of index

Then $x = k + 1$, so

$$E[X] = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^{k+1}}{k!}.$$

Taking λ outside,

$$E[X] = \lambda e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}.$$

Using the exponential series

$$e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!},$$

Taylor series: $e^{\lambda} = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!}$

we obtain

$$E[X] = \lambda e^{-\lambda} e^{\lambda} = \boxed{\lambda}.$$

(b) Derive the MLE of λ

Suppose we observe n independent observations

$$x_1, x_2, \dots, x_n.$$

Likelihood:

For each observation,

$$P(X_i = x_i) = \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}.$$

Independent samples: joint pmf = product

Since the observations are independent,

$$L(\lambda) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!}.$$

Therefore,

$$L(\lambda) = \frac{e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}.$$

Log-likelihood:

$$\ln(ab) = \ln a + \ln b, \ln(a^k) = k \ln a$$

$$\ell(\lambda) = \ln L(\lambda)$$

so

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^n x_i \right) \ln \lambda - \sum_{i=1}^n \ln(x_i!).$$

Derivative and critical point:

$$(\ln \lambda)' = \frac{1}{\lambda}; \text{ critical point: } \ell' = 0$$

$$\ell'(\lambda) = -n + \frac{\sum_{i=1}^n x_i}{\lambda}.$$

Set the derivative equal to zero:

$$-n + \frac{\sum_{i=1}^n x_i}{\lambda} = 0.$$

Hence,

$$n\lambda = \sum_{i=1}^n x_i$$

and therefore

$$\hat{\lambda}_{\text{MLE}} = \frac{1}{n} \sum_{i=1}^n x_i = \bar{x}.$$

Second derivative:

Second derivative test: $\ell'' < 0 \Rightarrow$
maximum

$$\ell''(\lambda) = -\frac{\sum_{i=1}^n x_i}{\lambda^2}.$$

For $\sum x_i > 0$,

$$\ell''(\lambda) < 0.$$

Therefore, the critical point is a **maximum**. Thus,

$$\hat{\lambda}_{\text{MLE}} = \bar{x}.$$

Final results:

$$E[X] = \lambda$$

and

$$\hat{\lambda}_{\text{MLE}} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i.$$